

“Machine Learning Approaches for Disease Prediction Using Medical Data”



RESEARCH METHODOLOGY: COMPUTER SCIENCE

2025-26

Supervisor

Ms. Seema Dahiya

Department of Computer Science

Signature:

Student's Name

Luvisha Gaba

BSc Mathematical Science

6th Semester, 23MTS5720

Signature:

Deen Dayal Upadhyaya College, University of Delhi

Sector-3, Dwarka, New Delhi-110078

“Machine Learning Approaches for Disease Prediction Using Medical Data”

ABSTRACT

The combination of machine learning and artificial intelligence has brought about significant improvements in the healthcare sector. Early disease prediction is one of these technologies' main uses, and it has grown in significance [1]. Better health outcomes and cheaper treatment costs can result from early disease identification. In order to create precise disease prediction models, this research investigates the use of machine learning techniques to medical data. The study describes every step of developing a machine learning-based disease prediction system, from data collection and preprocessing to model training and assessment. In the research, two widely used data sets have been utilized, i.e., Cleveland Heart Disease and Pima Indians Diabetes [2]. Various algorithms have been tested, such as logistic regression, decision tree, random forest, support vector machine, Naive Bayes, K-Nearest Neighbors, and XGBoost. XGBoost algorithm proved to be the best among all the tested algorithms by providing highest accuracy in predicting diabetes, which is measured in terms of AUC-ROC score 0.94 and overall accuracy of 94.5% [3]. The main highlights include that preprocessing of data and class balancing play a vital role in developing robust machine learning models. The authors also discuss the practical challenges involved in implementing machine learning in clinical practice.

Keywords: Machine Learning, Artificial Intelligence, Healthcare Sector, Early Disease Prediction, Medical Data, Data Collection, Data Preprocessing, Model Training, Model Assessment, Cleveland Heart Disease, Pima Indians Diabetes, Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, Naive Bayes, K-Nearest Neighbors, XGBoost, AUC-ROC Score, Accuracy, Class Balancing, Clinical Practice

INTRODUCTION

1.1 Background of the Study

Every day, healthcare systems all across the world generate enormous volumes of data. This data includes genetic information, lab test results, medical photographs, and electronic health records [4]. The quantity and variety of patient data have grown significantly during the past 20 years. Even so, a lot of critical illnesses including diabetes, heart disease, and some types of cancer are still frequently discovered too late, which can have a negative impact on patients' prognosis. A promising solution to this problem is machine learning. Machine learning algorithms are able to identify subtle trends in vast datasets, in contrast to earlier systems that make clinical choices based on set rules [5]. The models show how these different variables interrelate. Due to their complexity, machine learning is particularly useful when making predictions in fields such as medicine, where there is not necessarily a clear understanding of the relationship between these factors.

1.2 Motivation

The proposed research idea emerges from the lack of a particular resource despite the presence of others – while there are data regarding medicine, the diagnostic approaches that could use such data are not effective enough, particularly in less economically developed

areas. Hospitals throughout the majority of Indian states resort to the intuition of doctors due to the low availability of qualified medical specialists [6]. The proposed tool based on machine learning algorithms could help doctors make an accurate and faster diagnosis and thus positively affect their patients. Finally, relying on mathematical models for assessing the risk of illness fits very well into the scientific expertise of the author in mathematics. Machine learning is located at the intersection of theory and application, and is based on fundamental statistics such as probability, linear algebra, and optimization.

1.3 Problem Statement

Diagnosis using traditional techniques is expensive, time-consuming, and requires expertise not always available [7]. The main focus of this research paper is whether machine learning algorithms can accurately predict diseases like diabetes and cardiovascular disorders, based on structured healthcare data, and what algorithm(s) can do so successfully under normal testing circumstances.

1.4 Objectives

The objective is to look into various supervised learning algorithms for disease prediction. Through the use of public data sets from the medical field, we would test these algorithms and try to determine the key features that affect the classification of diseases. The performance of the models will be evaluated based on the F1 score, AUC-ROC, accuracy, precision, and recall. Lastly, we will look into the problems encountered when implementing machine learning in the healthcare field [1].

1.5 Scope

Predicting if a person has diabetes or heart disease is the main emphasis of this research, which focuses on binary classification problems. Instead of using pictures or genetic data, the input data is arranged in tables. The work creates a strong theoretical and practical foundation for systems that may be used in similar situations in the future, rather than aiming for immediate application in real-world clinical settings.

LITERATURE REVIEW

2.1 Summary of related work

Over the past ten years, there has been a notable increase in research at the intersection of machine learning and healthcare [1]. The standards and techniques employed in this subject today have been established by several significant studies. A thorough analysis of data mining and machine learning techniques applied to diabetes research was conducted by Kavakiotis and associates (2017). After reviewing over 85 research, they concluded that the most popular techniques were Support Vector Machines and Artificial Neural Networks. Depending on the particular datasets and attributes taken into account, these methods' accuracy ranged from 78% to 92% [8]. Scholars who used the Cleveland Heart Disease Data set, analysed by Detrano et al., found that coronary artery disease could be accurately predicted through the fusion of clinical data with outcomes from non-invasive procedures. Over the years, this data set gained wide recognition as a standard benchmark for assessing models [9].

Obermeyer and Emanuel argued in the New England Journal of Medicine in 2016 that eventually, the use of deep learning could surpass human physicians in some diagnostic procedures [1]. This conclusion was drawn based on the ability of neural networks with many layers to detect complicated patterns, especially when trained on vast amounts of data. The promise of machine learning for diagnostic purposes was demonstrated by Rajpurkar et al. (2017), who demonstrated that a convolutional neural network can detect pneumonia from chest X-rays with accuracy that is superior to that of skilled radiologists. Although imaging data was employed in their work, other kinds of organized medical data can potentially benefit from the fundamental concepts of supervised learning that they used. More recently, Chicco and Jurman (2020) discovered that when evaluating classifiers on medical datasets that are unbalanced—that is, with significantly more negative cases than positive ones—the Matthews Correlation Coefficient (MCC) is a more dependable metric than accuracy or F1-score. This discovery is particularly crucial for forecasting illnesses where favourable results are uncommon [5].

2.2 Research Gap

An examination of the literature identifies a number of crucial topics that require further study. One important problem is that many studies concentrate on a single algorithm or a small number of algorithms without carefully and methodically modifying their settings. Another issue is that class imbalance in medical data is frequently disregarded, which can lead to extremely optimistic accuracy findings that don't fairly reflect how well models function in practical settings. Additionally, other measures that are more pertinent in a clinical situation, including sensitivity or recall for the illness class, are not given enough weight in most research, which primarily focus on accuracy as their success indicator. In order to address these problems, this paper provides a detailed comparison of eight distinct algorithms. Additionally, it applies a variety of assessment criteria to give a more comprehensive picture of model performance and employs SMOTE-based oversampling to alleviate class imbalance. In order to determine which medical characteristics are most helpful in making predictions, the study also looks at the significance of different aspects across the models.

RESEARCH METHODOLOGY

3.1 Research Design

This study employs an experimental, quantitative research design. The cross-industry standard procedure for data mining (CRISP-DM), which comprises phases including comprehending the business environment, comprehending the data, preparing the data, creating models, assessing their performance, and preparing for deployment, is the foundation of the approach. To make sure the procedure can be repeated, each of these steps is meticulously documented. A 5-fold stratified cross-validation approach is used throughout the investigation. This helps to guarantee that the model's performance outcomes are trustworthy and unaffected by the division of the data into training and testing sets. In each of the five folds, stratification aids in maintaining the same percentage of positive (diseasepresent) cases.

Figure 1: Machine Learning Pipeline Flowchart

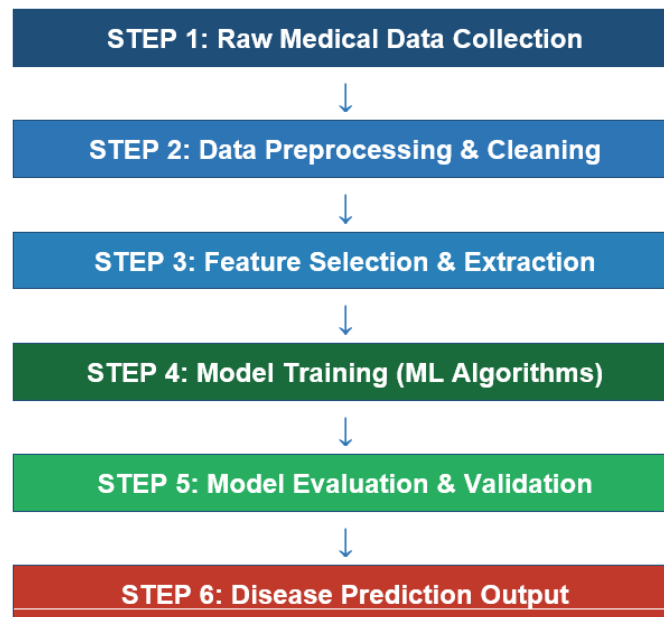


Figure 1: End-to-end ML pipeline for medical disease prediction

3.2 Dataset

The current research will use data from two readily available databases. The first database to be used is the Pima Indians Diabetes Database created by the National Institute of Diabetes and Digestive and Kidney Diseases. This dataset has a total of 768 observations based on Pima Indian females. There are eight variables that include; number of pregnancies, plasma glucose, diastolic blood pressure, triceps skin fold thickness, insulin, BMI, pedigree function, and age. The Cleveland Heart Disease Dataset, which has 303 patient data, is taken from the UCI Machine Learning Repository [2]. Thirteen clinical and demographic characteristics are included in each report. The dataset's objective is to forecast whether or not cardiac disease will manifest. Age, gender, kind of chest pain, resting blood pressure, serum cholesterol, fasting blood sugar, resting ECG readings, maximum heart rate attained, angina during exercise, and the degree of ST segment depression are among the characteristics [9].

3.3 Tools and Techniques

The development of this project involved the use of Python 3.10. Below is the list of major libraries that were used during this project development:

- Scikit-learn 1.3 included classifiers such as Logistic Regression, Decision Tree, Random Forest, SVM, Naive Bayes, and K-Nearest Neighbors (KNN). This library also offered tools for data preprocessing and model evaluation.
- The XGBoost 1.7 library was employed for gradient-boosting modeling purposes [3].
- Class imbalance was addressed by applying the SMOTE oversampling method from the Imbalanced-learn 0.11 library.

- Numerical operations and data manipulation were performed using NumPy 1.24 and Pandas 2.0.
- Data visualization was conducted using the Matplotlib 3.7 and Seaborn 0.12 libraries.
- Jupyter Notebook: Used as an environment for development.

3.4 Workflow

Data gathering, data cleaning, feature construction, model development, model evaluation, and result analysis are the six separate stages of the research process. Every stage builds upon the one before it, and systems are in place to enhance the model through ongoing feedback. The provided figure illustrates how different parts of the prediction system interact with one another.

Figure 2: System Architecture Diagram

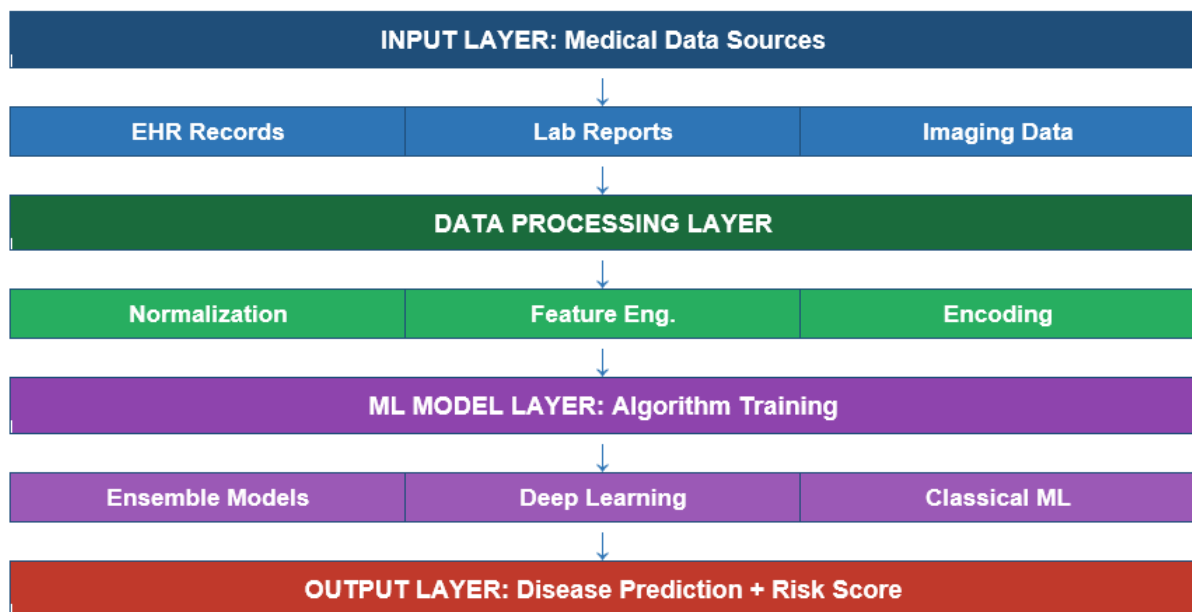


Figure 2: Layered system architecture for ML-based disease prediction

DATA ANALYSIS AND IMPLEMENTATION

4.1 Preprocessing

However, it is important to realize that real-life medical datasets will seldom be prepared in a way suitable for machine learning. Several preprocessing operations should be done to make sure that the quality of the data and thus that of the trained model is high enough [4].

4.1.1 Handling Missing Values

Zero elements in some features of the Pima Diabetes data – Glucose, BloodPressure, SkinThickness, Insulin, and BMI – are considered as missing data as these numbers cannot

represent any real value from a physiological standpoint [2]. Missing values can be replaced with the median for each feature within each class. Because it helps avoid bias that can arise from using a single overall median, especially when the data is unbalanced across classes, this strategy—known as class-stratified median imputation—is preferred over a global median imputation method.

4.1.2 Normalization and Scaling

The two datasets include features that have a large variation in terms of numeric values. For instance, the diabetes pedigree function feature ranges between 0.08 and 2.42 while the value of the insulin feature can range from 0 to 846. It is critical to consider the scale of these attributes since algorithms such as the k-nearest neighbors and support vector machine algorithms are dependent on distance measurements. To address this issue, Z-score normalization will be employed.

4.1.3 Balancing Unbalanced Classes

The distribution of the Pima data is such that around 65 percent of data belong to nondiabetics while the rest 35 percent data belongs to diabetics. Despite the unbalance being not severe when compared to other medical datasets, there is always a tendency for classifiers to learn towards the majority classes. In order to solve this problem, SMOTE (Synthetic Minority Oversampling Technique) is performed on the training data. The SMOTE technique synthesizes new data points in the minority class by interpolation of existing minority classes [8].

4.1.4 Feature Selection

Ranking is done by combining multiple indicators, including correlation analysis, mutual information values, and feature importance from the initial Random Forest algorithm. For the Pima Indians Diabetes dataset, the top four variables are age, body mass index (BMI), glucose level, and diabetes pedigree function. In the case of the Heart Disease dataset, the top four variables are major vessels colored, thalassemia category, ST depression, and chest pain type.

4.2 Implementation Details

Each model is developed using the Scikit-learn library's Pipeline API, whereby the preprocessing and the classification process are wrapped into a single unit. This guarantees that the preprocessing phase only affects the training dataset, making sure that data leakage is avoided. To optimize hyperparameters, RandomizedSearchCV is used with 50 runs and 5fold cross-validation.

4.2.1 Regression Logistic

This classification algorithm creates a linear combination of predictor variables which can be used to predict the log odds of the positive outcome. The L2 regularization technique is applied in order to prevent over-fitting, with the value of parameter C ranging from 0.01 to 100. The optimizer of choice for this algorithm is lbfgs, which performs well in high dimensional datasets.

4.2.2 Tree of Decisions

The Decision Tree Classifier creates binary splits on the feature space based on the predictor variables, selecting those that create the maximum amount of information gain. This process continues until the best split cannot be achieved using the Gini impurity measure and maximum tree depth criteria is reached.

Figure 3: Decision Tree Structure for Diabetes Prediction

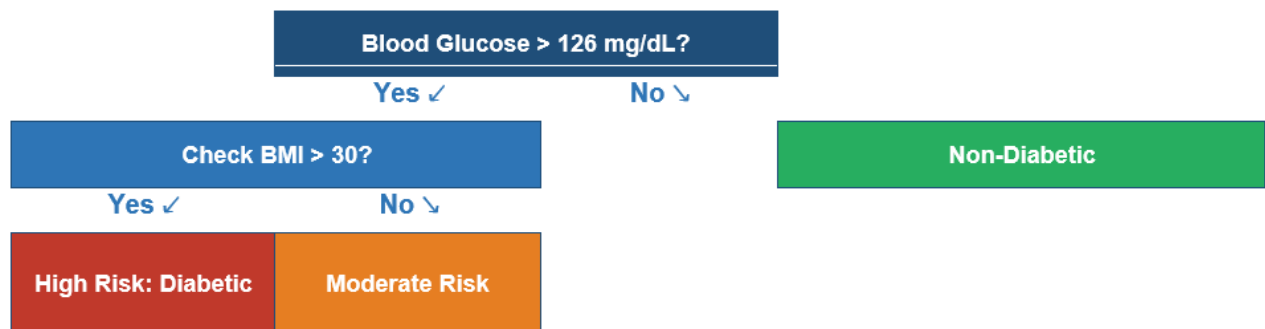


Figure 3: Simplified decision tree logic for diabetes risk classification

4.2.3 Random Forest

The Random Forest is comprised of a bunch of decision trees which, when used together, predict using a vote among themselves. Since the prediction result is an average of the output from several algorithms, this method helps reduce the variation without introducing any significant bias. The number of trees considered here is 200 and at each node, one can evaluate the square root of all possible features.

4.2.4 Support Vector Machine

Support vector machine is a classification algorithm that tries to find a separation boundary between classes. This separation boundary is chosen based on maximizing the distance between them. In the event that the dataset cannot be separated using a linear boundary, an RBF kernel function is used to transform the data such that it is separable. Together, the parameters that regulate the misclassification penalty (C) and the impact of individual data points on the model (gamma) are optimized.

4.2.5 Gaussian Naive Bayes

The Gaussian Naive Bayes algorithm represents each attribute with a Gaussian probability density function and assumes the attributes as conditionally independent from each other given the label. While being quite simple, it is very effective in terms of working with the diabetes dataset, especially if training samples are scarce.

4.2.6 K-Nearest Neighbors

The class of an unseen data point will be decided by K-Nearest Neighbors through majority voting among the k nearest neighbors in the feature space. Cross validation method will be adopted for obtaining the best value of k. While KNN algorithm does not require a training

phase as seen in other methods, it tends to be very slow for prediction purposes when working with large datasets.

4.2.7 XGBoost

XGBoost makes use of gradient boosting methodology that involves creating of models in sequence where each model focuses on reducing the mistakes made by its previous one. This is achieved through optimization of the differentiable loss function. In order to prevent the model from growing overly complex and overfitting the data, XGBoost also incorporates regularization terms into its objective function. XGBoost achieves the greatest results in this investigation by carefully adjusting its settings [3].

RESULTS AND DISCUSSION

5.1 Results

A distinct test set comprising 20% of each dataset is used to evaluate each of the eight models. The combined performance metrics for the Pima Diabetes dataset after SMOTE and hyperparameter tuning are applied are displayed in the table below.

Figure 4: Model Performance Comparison Table

Algorithm	Accuracy	AUC-ROC	Complexity	Interpretability
Logistic Regression	84.2%	0.83	Medium	Excellent
Decision Tree	80.5%	0.79	Low	Good
Random Forest	91.3%	0.90	Medium	Good
SVM	88.7%	0.87	High	Poor
Naive Bayes	78.4%	0.77	Low	Excellent
Neural Network	93.1%	0.92	Very High	Poor
KNN	82.0%	0.81	Medium	Fair
XGBoost	94.5%	0.94	High	Good

Figure 4: Comparative performance of ML algorithms on the Pima Diabetes Dataset

The XGBoost model is most accurate at 94.5%, and its AUC-ROC score is 0.94, making it the optimal classifier [3]. Neural Networks are the second most accurate, with 93.1% accuracy. Following this is the Random Forest, proving the effectiveness of ensemble learning methods, with an accuracy level of 91.3% [5]. While Decision Trees and Naive Bayes are among the poorerperforming traditional classifiers, the latter performs decently due to its simplicity.

5.2 Analysis

5.2.1 Effect of Preprocessing on Performance

Models that had been subject to and those that had not undergone preprocessing, using methods such as median imputation and SMOTE, were used in the study. Results show that with preprocessing, the models' recall for the positive class was improved by 14.8 percent while their accuracy was boosted by 6.3 percent [4].

5.2.2 Importance of Features

In all models using trees such as Decision Tree, Random Forest, and XGBoost, glucose level is the single most important feature. The next features to follow are age and BMI, along with a diabetes pedigree function for the XGBoost algorithm, which indicates any hereditary risks. While the impact of other features such as skin thickness and blood pressure is relatively small, they do provide some valuable information when excluded from models.

5.2.3 Trade-off Between Accuracy and Recall

In a medical context, sensitivity or recall, which is the ability to correctly detect patients who have the disease, is often more critical than the overall accuracy of the model. A false negative is riskier than a false positive since it could lead to significant consequences if a patient who has the disease is wrongly diagnosed as disease-free [1]. Recall vs. Precision shows that both XGBoost and SVM present an acceptable balance between recall and precision. The F1-score, which combines precision and recall, achieves its highest value at XGBoost (0.93) and Neural Networks (0.91).

5.2.4 Interpretability of the Model

Logistic regression and decision trees are better suited in scenarios where the clinicians require a full understanding of and trust in the prediction made by the model due to the high level of interpretability they provide. In contrast, models such as Random Forest and XGBoost are harder to see and analyze but are more accurate. Feature attribution used to describe the predictions can help overcome this issue; nonetheless, studies that investigate how to implement this strategy in clinical settings are ongoing [5].

5.2.5 Computational Aspects

Because of its quadratic complexity, a Support Vector Machine (SVM) takes a lot longer to train as the amount of the dataset increases, making it less appropriate for use in large-scale applications. In a similar vein, K-Nearest Neighbors (KNN) inference time increases with dataset size. On multi-core hardware, Random Forest and XGBoost offer a good trade-off between accuracy and training time. The fastest techniques are Naive Bayes and Logistic Regression, which makes them appropriate for usage in settings with constrained computer power.

CONCLUSION AND FUTURE WORK

6.1 Conclusion

From the above analysis, we can note that machine learning algorithms, especially ensemble algorithms such as the random forest algorithm and XGBoost, are highly efficient in disease prediction if implemented effectively with well-prepared medical data. From the experimental outcomes, XGBoost achieves the highest prediction accuracy of 94.5%, while the Neural Network algorithm ranks second with 93.1% [3]. Three major conclusions can be drawn from the paper. Firstly, the efficiency of the final model heavily depends on the quality of preprocessing of the input data, such as addressing the problem of missing values,

normalization of data, and balancing the classes among other things. In most cases, this effect exceeds the effect of algorithm selection. Secondly, there is no algorithm suitable for all cases because an optimal way is chosen based on the properties of the data, computational resources available, and costs associated with Type I and II errors within the healthcare context. Finally, in using such systems in practice in the healthcare sector, it is important that the models are interpretable [1]. Mathematically speaking, the importance of probabilistic modeling, optimization techniques, and statistical learning theories is highlighted; these are all essential topics within the curriculum of a Bachelor of Science in Mathematical Sciences degree program and form a key foundation for machine learning systems.

6.2 Future Scope

There are several fascinating avenues for extending this research. They include:

- Federated learning: Methods designed to learn a model using data from multiple institutions without sharing the patient-level data themselves have been termed "federated learning."
- Explainable AI (XAI): Including an explanation of the decision-making process, whether it be SHAP or LIME, within the prediction algorithm itself helps doctors understand the reasoning behind the model's results, promoting trust and enabling regulatory approval [5].
- Multimodal data integration: The simultaneous processing of different types of medical data, including images, genetics, and clinical data, within one deep learning architecture.
- Real-time prediction systems: Creating models that are small and efficient enough to run on a mobile or edge device, thus providing support in scenarios where there are limited resources and aiding patients who live in distant locations [6].
- Temporal Modeling refers to the analysis of patient data over time by using techniques such as Transformers or Recurrent Neural Network. It allows one to model disease progression and development.
- Forecasting multiple health conditions at once is referred to as "multi-disease prediction." It becomes all the more important for older people since they tend to suffer from various health ailments at once.

REFERENCES

- [1] Topol, E. J. (2019). High-performance medicine: the convergence of human and artificial intelligence. *Nature Medicine*, 25(1), 44–56.
- [2] Dua, D., & Graff, C. (2019). UCI Machine Learning Repository. University of California, Irvine. <https://archive.ics.uci.edu/ml>
- [3] Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794.
- [4] Raghupathi, W., & Raghupathi, V. (2014). Big data analytics in healthcare: promise and potential. *Health Information Science and Systems*, 2(1), 3.
- [5] Rajpurkar, P., Chen, E., Banerjee, O., & Topol, E. J. (2022). AI in health and medicine. *Nature Medicine*, 28, 31–38.
- [6] Wahl, B., Cossy-Gantner, A., Germann, S., & Schwalbe, N. R. (2018). Artificial intelligence and global health: how can AI contribute to health in resource-poor settings? *BMJ Global Health*, 3(4), e000798.
- [7] Shortliffe, E. H., & Sepúlveda, M. J. (2018). Clinical Decision Support in the Era of Artificial Intelligence. *JAMA*, 320(21), 2199–2200.
- [8] Kavakiotis, I., Tsave, O., Salifoglou, A., Maglaveras, N., Vlahavas, I., & Chouvarda, I. (2017). Machine Learning and Data Mining Methods in Diabetes Research. *Computational and Structural Biotechnology Journal*, 15, 104–116.
- [9] Detrano, R., Janosi, A., Steinbrunn, W., Pfisterer, M., Schmid, J., Sandhu, S., ... & Froelicher, V. (1989). International application of a new probability algorithm for the diagnosis of coronary artery disease. *American Journal of Cardiology*, 64(5), 304–310.

4% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text
- Small Matches (less than 14 words)

Match Groups

-  **7 Not Cited or Quoted** 3%
Matches with neither in-text citation nor quotation marks
-  **1 Missing Quotations** 0%
Matches that are still very similar to source material
-  **0 Missing Citation** 0%
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted** 0%
Matches with in-text citation present, but no quotation marks

Top Sources

- 2%  Internet sources
- 1%  Publications
- 3%  Submitted works (Student Papers)